Guarding Walls Contiguously

An interactive exploration of the Contiguous Art Gallery Problem

Project proposal • Ilias Akajou (520466) and Jomaa Abdulrahman (583629) • October 19, 2026

Topic	The Contiguous Art Gallery Problem.
Main article	S. de Berg, J. Conradi, I. van der Hoog, F. Staals, *Simpler and Faster Contiguous Art Gallery* [5]. We will also use the SoCG 2025 paper [1] and the $\Theta(n \log n)$ result [6].
Team	Ilias Akajou (520466) and Jomaa Abdulrahman (583629).
Deliverable	An interactive webpage on GitHub Pages, with our report and our JavaScript applets.

1 The problem

Let P be a simple polygon with n vertices, and let ∂P be its boundary. A point $g \in P$ *sees* a point q if the segment gq is inside P. A *contiguous guard* is a point $g \in P$ together with a connected part C of ∂P, called its *chain*, such that g sees every point of C. The *Contiguous Art Gallery Problem* asks for the minimum number k of contiguous guards needed so that their chains cover all of ∂P. We can assume that the chains do not overlap, so they form a partition of the boundary. Figure 1 shows an example where $k = 2$.

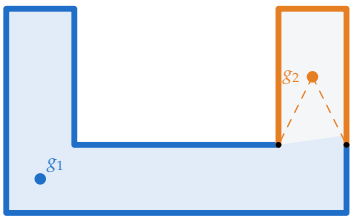


Figure 1. An example with two contiguous guards. Each guard sees the whole chain of its colour, and the shaded area is what g_1 sees (its visibility polygon). P is not star-shaped, so one guard is not enough and $k = 2$.

An easy way to think about it: each guard is a security camera that turns and films one part of the wall, without skipping anything. The problem was proposed by T. Shermer at the open problem session of CCCG 2024.

We chose this topic mainly because of one surprising fact. In the classical art gallery problem, where we want the minimum number of point guards that see the whole polygon, the problem is NP-hard [8] and even $\exists\mathbb{R}$-complete [9]. Guarding only the boundary is also $\exists\mathbb{R}$-hard [10]. So we first expected that adding a contiguity constraint would make the problem even harder, but it is actually the opposite: the contiguous version can be solved in polynomial time [1], and even in $\Theta(n \log n)$ time [6]. We wanted to understand why.

2 Previous work

This problem was solved very quickly: in about one year, it went from an open question to an optimal algorithm. Three groups found polynomial algorithms independently [2, 3, 4], and their results were merged into one paper at SoCG 2025 [1]. Each group used a different idea: [2] repeats greedy traversals of the boundary, [3] builds $O(n^4)$ candidate guards and combines them with a combinatorial argument, and [4] reduces the problem to the *Analytic Arc Cover Problem* and uses real algebraic geometry. These proofs are quite long (48, 22 and 21 pages).

A few months later, de Berg et al. [5] gave a much shorter proof that is almost self-contained. This is the article we will focus on, because it is short enough for us to understand it in detail. The same authors then found an optimal algorithm with a matching lower bound [6]. Table 1 summarises these results.

Year	Reference	Main idea	Running time
2024–25	[2, 3, 4] → [1] (SoCG)	three independent approaches	$O(kn^5 \log n)$, $O(n^6 \log n)$
2025	[5] (**our focus**)	greedy + $O(n^2)$ candidate starting points	$O(kn^2 \log^2 n)$
2025	[6]	optimal algorithm + lower bound	$\Theta(n \log n)$

Table 1. Algorithms for the Contiguous Art Gallery Problem.

3 Proposal

We want to make an interactive webpage that explains the problem and the algorithm of [5] step by step. It will have four parts.

1. **The problem.** We introduce the problem with interactive figures and compare it to the classical art gallery problem [7], which is NP-hard [8] and even $\exists\mathbb{R}$-complete [9].
2. **The algorithm of [5].** This is the most important part of our project. Using visual examples, we want to explain:
   - the visibility core, and why one contiguous guard is enough if and only if P is star-shaped;

- the greedy algorithm: starting from any point of ∂P it uses at most $k+1$ guards, and exactly k if it starts at the beginning of a chain of an optimal solution;
- how the arrangement of the lines through the edges of P, together with the difference between good and bad guards, gives $O(n^2)$ candidate starting points, at least one of which is optimal;
- how one greedy step is computed with shortest paths and ray shooting [11, 12], which gives a total running time of $O(kn^2 \log^2 n)$;
- more briefly, how [6] improves this to the optimal $\Theta(n \log n)$.

3. **A game.** To make the page more fun, we will add a small game. The page generates a random simple polygon, and the player places guards and chooses, for each guard, the part of the boundary it watches. The game checks that each guard really sees its whole chain and that all the chains together cover the boundary. Then it compares the player's number of guards with the optimal k computed by our implementation: if they are equal the player wins, otherwise we show an optimal solution. There will also be a sandbox mode where users can draw their own polygons, for example to try the examples from parts 1 and 2.
4. **The report.** The report will be directly on the webpage instead of a separate PDF. It will contain the definitions, the main results and proof sketches, in the same style as the rest of the page so that it is easy to read.

4 Plan

First, we will read the articles carefully, starting with [5]. Then we will implement the solver (visibility polygons, maximal chain seen by a guard, greedy algorithm and candidate starting points) and build the game on top of it. Having the solver early will help us create our own examples and figures for the rest of the page. After that, we will do the other parts one by one. Everything will run in the browser, in JavaScript with React for the interface, so the page can be hosted on GitHub Pages. We will also write an AI journal and a short document explaining how we shared the work.
We will divide the work as follows :
Ilias Akajou will work on part one, Jomaa Abdulrahman will work on part two and both will work on part three and four.

We expect two main difficulties: generating random simple polygons for the game, and handling bad guards in the algorithm, which seems to be the most technical part of [5]. We will not implement the fast data structures of [5] and [6]: simpler polynomial-time versions should be enough for polygons with a few dozen vertices.

References

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[3] A. Biniaz, P. Bose, A. Maheshwari, J. S. B. Mitchell, S. Odak, V. Polishchuk, and T. C. Shermer. Contiguous boundary guarding. arXiv:2412.15053, 2024. Merged into [1].

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[5] S. de Berg, J. Conradi, I. van der Hoog, and F. Staals. Simpler and faster contiguous art gallery. arXiv:2508.09734, 2025. Described by its authors as a preliminary version of [6]; it contains the $O(kn^2 \log^2 n)$-time algorithm studied in this project, which differs from the final algorithm of [6].

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